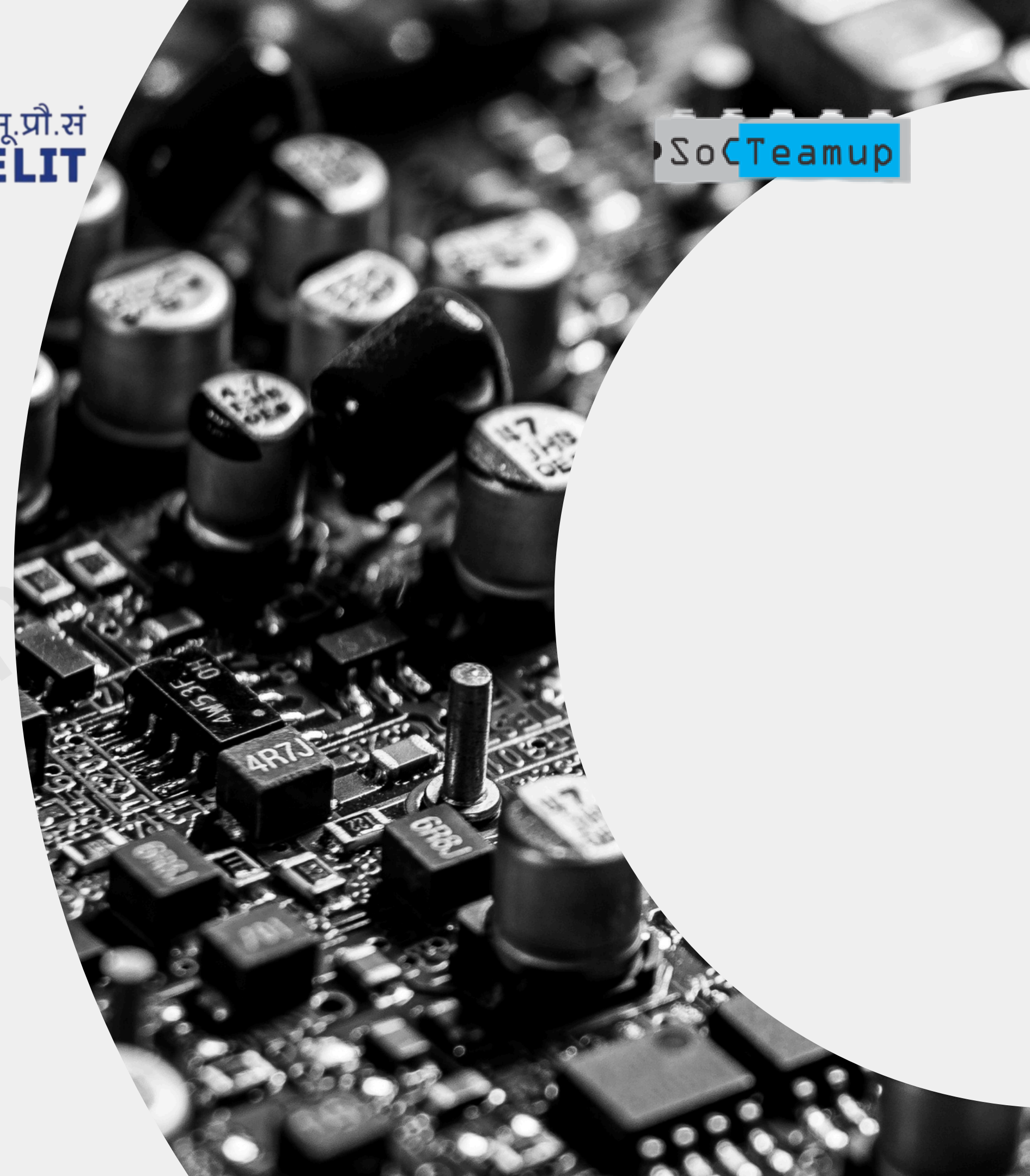


Understanding Setup time, Hold time and Clock skew in Synthesis





Outline

Why timing matters in digital circuits?
Flip-flop behaviour and timing
What is setup time?
What is hold time?
Setup vs hold constraints
Clock-to-Q delay
Timing diagram walkthrough
Clock period and slack
What is clock skew?
Positive vs negative skew
Impact of skew on setup/hold
Setup/hold violations – causes
How synthesis handles these constraints
Improving timing in RTL
Summary and Best Practices

➔ Why Timing Matters ?

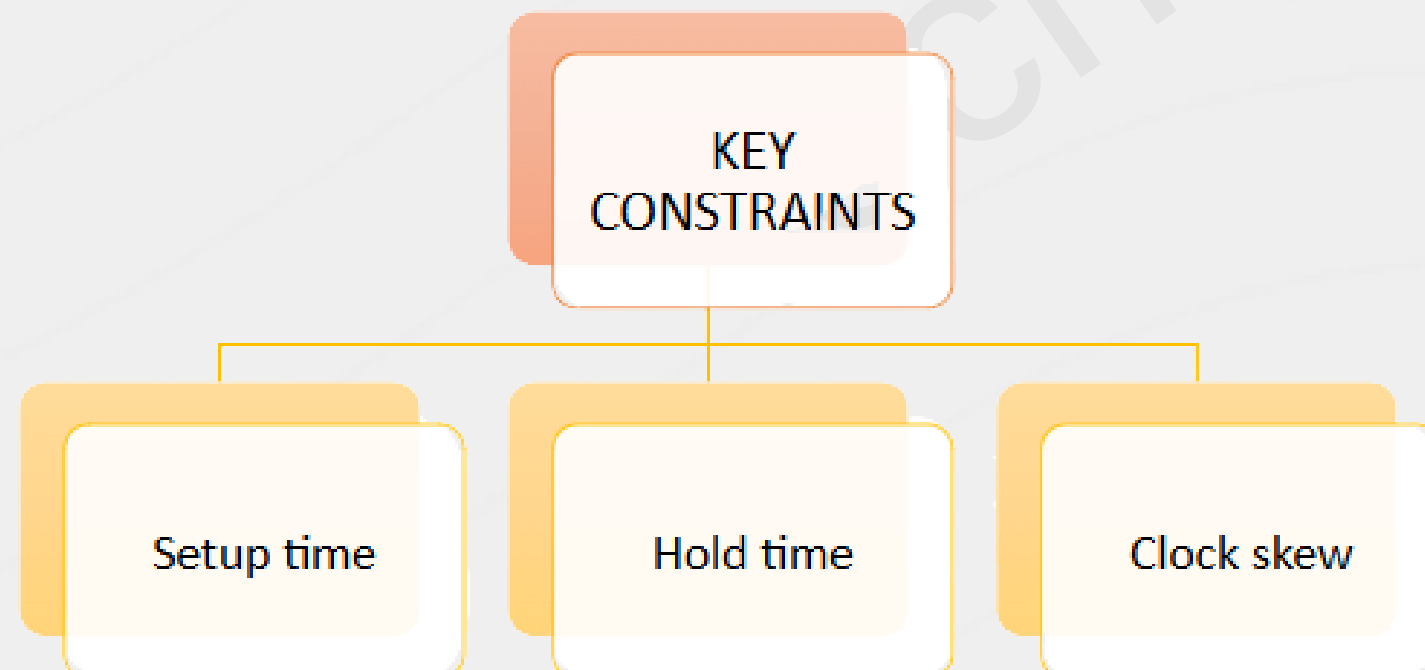
Timing matters because digital circuits must complete computations within strict clock periods.

Key Reasons

Meets Clock Frequency: Paths longer than clock cycle cause setup violations, dropping max speed.

Prevents Metastability: Proper hold/setup times avoid flip-flop glitches and data corruption.

Ensures Reliability: Violated timing creates unpredictable behavior or functional failures.



What is Setup Time?

Setup time = Minimum time before clock edge when data must be stable

Example:

- Data must be stable 20ps before rising edge
- If not, flip-flop may sample wrong value → setup violation

Constraint: Arrival time + setup time $\leq$ clock edge

What is **Hold Time**?

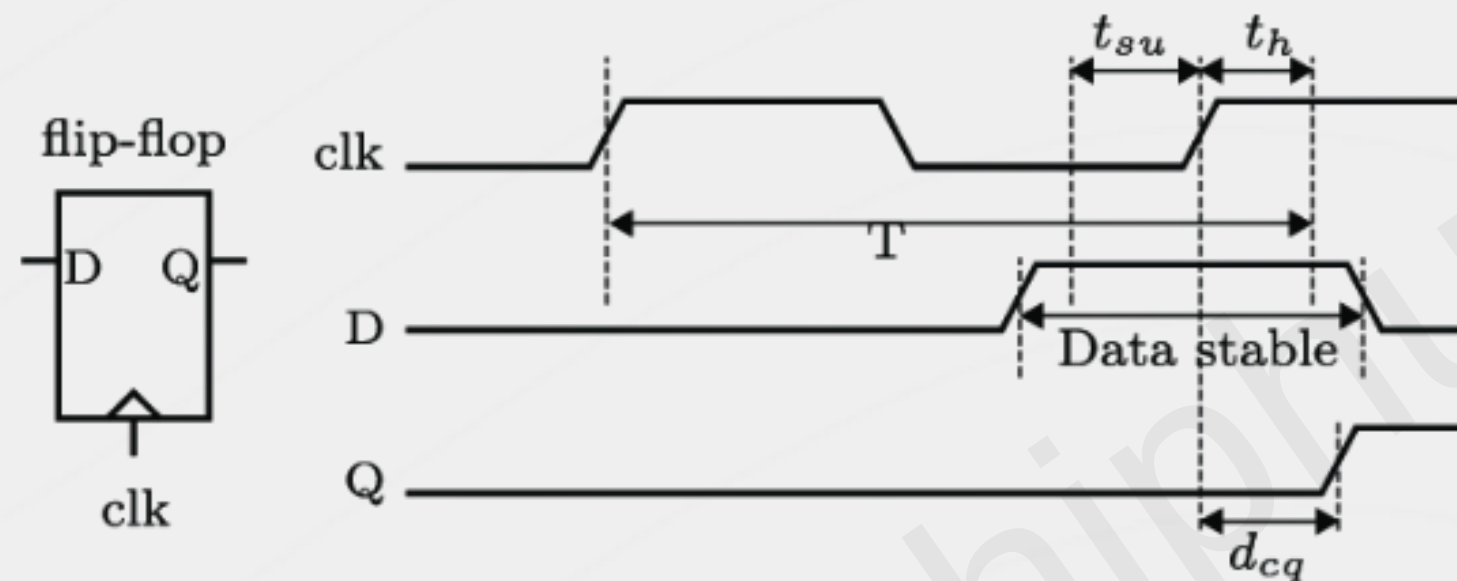
Hold time = Minimum time after clock edge during which data must remain stable

Example:

- If data changes too soon after clock edge → hold violation

Constraint: Arrival time $\geq$ clock edge + hold time

➔ Setup vs Hold – Visualized



- Setup(t_{su}) → input must arrive early enough
 - Hold(t_h) → input must not change too soon after
- Both are critical for reliable flip-flop sampling

Clock-to-Q Delay (D_{cq})

D_{cq} = Time from clock edge to when Q output becomes valid

Affects:

- Launch time of data
- Next flip-flop setup timing

Used in timing equations: $\text{Arrival} = \text{Launch} + D_{cq} + \text{Logic Delay}$

tsu : setup time

th : hold time

T : clock period

dcq : clock to q delay

Clock Period and Slack

Clock Period = Time between two rising edges

Slack = Time margin between required and actual arrival

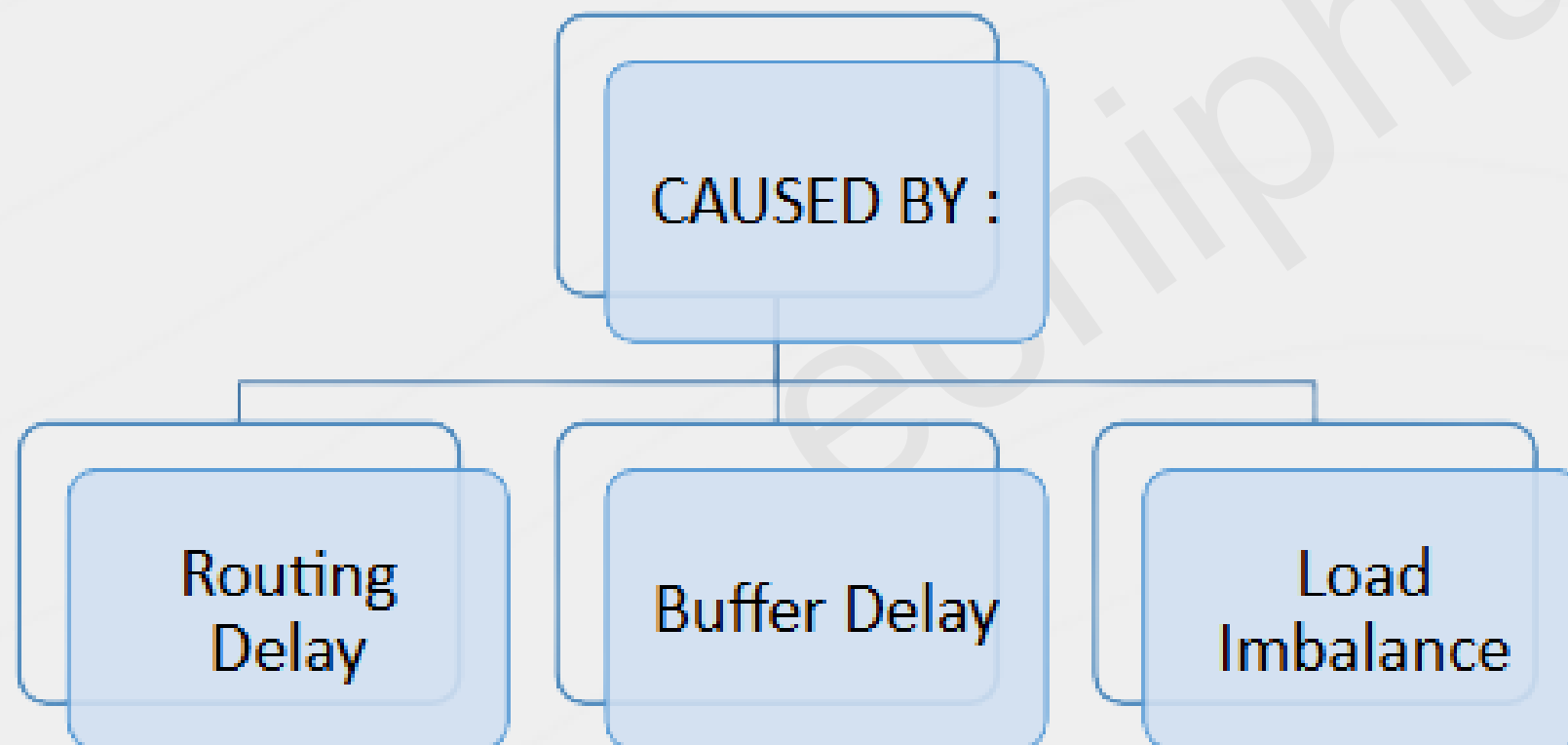
Positive Slack = Good

Negative Slack = Violation (Setup & Hold)

Synthesis tools aim to maximize slack

➔ What is Clock Skew?

Clock Skew = Difference in arrival time of clock at two flip-flops



➔ Setup and Hold Violations – Causes

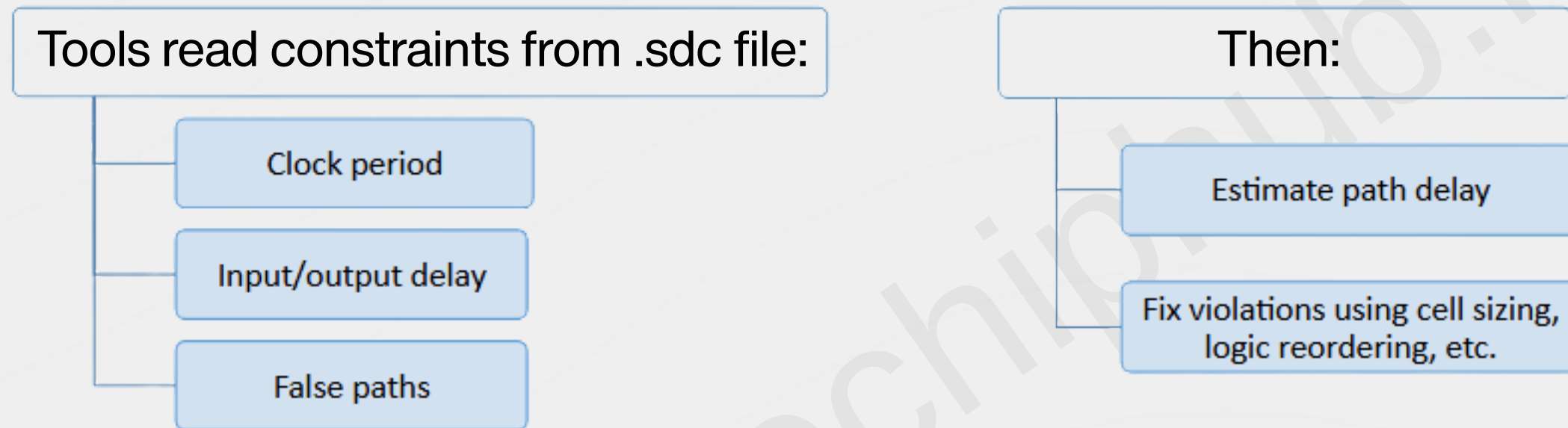
CAUSES

- Long data paths (combinational delay)
- Short paths with low delay (causing hold violation)
- Poor reset or clock gating logic
- Clock skew and jitter

FIX

- Pipeline stages
- Logic balancing
- Buffer insertion

➔ How Synthesis Tools Handle Timing



Improving Timing in RTL

Avoid long combinational chains

Register outputs early

Balance fan-out and logic levels

Don't infer unintended latches

Use pipeline registers in high-speed paths

➔ Summary and Best Practices

Setup time → data arrives before clock edge

Hold time → data stays stable after clock edge

Clock skew → shifts effective timing window

Timing violations → logic failure

Synthesis considers all of this using constraints

Understand these to write timing-friendly RTL!

Thank You

echiphu

